# FIITJEE

# ALL INDIA TEST SERIES

## FULL TEST – VIII

## JEE (Advanced)-2019

**PAPER – 2**

# ANSWERS, HINTS & SOLUTIONS

## *Physics*

**PART – I**

**SECTION – A**

1. AD

Sol. $y_1 = a\cos\omega_1 t$ and $y_2 = a\cos(\pi + \omega_2 t)$

For same phase $\omega_1 t = \pi + \omega_2 t$

$$t = \frac{\pi}{\omega_1 - \omega_2} = \frac{\pi}{\frac{2\pi}{3} - \frac{2\pi}{7}} = \frac{21}{8}s$$

Or

$$t = \frac{3\pi}{\omega_1 - \omega_2} = \frac{3\pi}{\frac{2\pi}{3} - \frac{2\pi}{7}} = \frac{63}{8}s$$

2. AB

Sol. $A_1$ will shrink : $B_2$ will shrink

$A_1$ will shrink : $A_2$ will expand

3. ABCD

Sol. Particle velocity $v_p = -v\left(\frac{dy}{dx}\right)$

$v$ is the wave velocity.

$\frac{dy}{dx}$ is the slope.

At point S, slope is zero, there force $V_p$ at S is zero.

At point T, slope is (+)*ve*, there fore $V_p$ will be along *–ve x* direction

Excess pressure $dP = -B.\frac{dy}{dx}$

At point S, slope is zero

$\Rightarrow$ dP = 0.

At point R, slope is *–ve*

$\Rightarrow$ dP is (+) ve i.e., particles located near C are under compression.

4. AC

Sol. If the image is real and magnified means object is between *f* and 2*f*.

When lens immersed in water focal length, $f_1 = \frac{(\mu - 1)}{\left(\frac{\mu}{\mu_r} - 1\right)} f = 4f$

Now object is between pole and focus so image is virtual and magnified.

5. BC

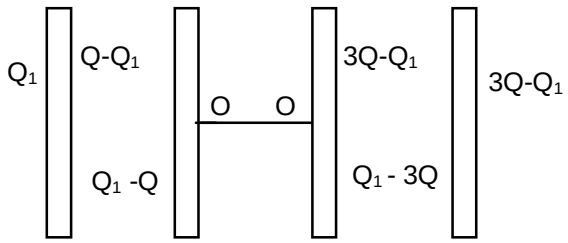
Sol. As $\lambda_0 = \frac{hc}{eV}$ and as *V* increases. $\lambda_0$ decreases but characteristic wavelengths do not change so interval between $\lambda_{k\alpha}$ & $\lambda_0$ increases and the same for the interval between $\lambda_{k\beta}$ & $\lambda_0$

6. BCD

Sol. Apply Faraday law for direction of induced current.

7. A

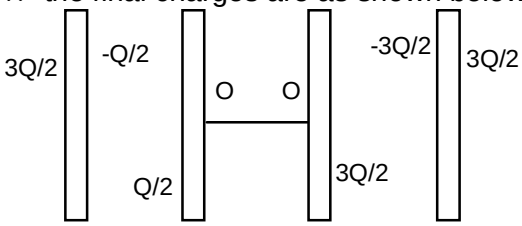
Sol.




Equating net electric field equal to zero in left most plate.

$3Q - Q_1 = Q_1 \Rightarrow Q_1 = \frac{3Q}{2}$.

$\therefore$ the final charges are as shown below.




8. B

Sol. Force $F = (-)\frac{dU}{dx}$

9. C

Sol. $I = \frac{2+3-5+4+6}{2+3+5+4+6} = \frac{1}{2}A$

$V_D = 2 - \frac{1}{2}\times 2 = 1V$ ; $V_C = V_D + 3 - \frac{1}{2}\times 3 = 2.5\,V$

$V_B = V_C - 5 - \frac{1}{2}\times 5 = -5\,V$ ; $V_A = V_B + 4 - \frac{1}{2}\times 4 = -3V$

10. D

Sol. $v^2 = 144 - 9x^2$

$\omega^2 = 9 \quad \Rightarrow \quad \frac{2\pi}{T} = 3 \Rightarrow T = \frac{2\pi}{3}$

when v = 0, $x = \sqrt{\frac{144}{9}} = \frac{12}{3} = 4$ unit

Acceleration = $v\frac{dv}{dx} = -9x$

Potential energy = $\frac{1}{2}m\omega^2x^2$ = 18 units

## SECTION – D

11. 00003.00

Sol. $g = \frac{GM}{R^2} = \frac{G\frac{4\pi}{3}R^3\rho}{R^2} = \frac{4\pi G}{3}\rho R$

$\frac{\sqrt{6}}{11} = \frac{g_1}{g} = \frac{\rho_1 R_1}{\rho R} = \frac{2}{3}\frac{R_1}{R}$

$\frac{R_1}{R} = \frac{3\sqrt{6}}{22}$

$\frac{V_1}{V} = \sqrt{\frac{M_1}{M}\frac{R}{R_1}} = \sqrt{\frac{\rho_1}{\rho}\frac{R_1^3}{R^3}\frac{R}{R_1}} = \sqrt{\frac{2}{3}\left(\frac{3\sqrt{6}}{22}\right)^2} = \frac{3}{22}\sqrt{\frac{2}{3}\times 6}$

$v_1 = 11\times\frac{3}{22}\times 2 = 3\ kms^{-1}$

12. 00002.72

Sol. Loss in KE = $K_f - K_i$

$K_f = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$

$K_i = \frac{1}{2}(0.08)(10)^2 + \frac{1}{2}(0.08)(6)^2$

Apply conservation of momentum and angular momentum to get $V_{cm}$ and $\omega$.

13. 00002.98

Sol. $T_{sp} = -K\ell^2\theta = \left(\frac{m\ell^2}{3}\right)\alpha$

$$\alpha = -\left(\frac{3K}{m}\right)\theta$$

$$T = 2\pi\sqrt{\frac{m}{3K}} \Rightarrow f = \frac{1}{T}.$$

14. 00000.25

Sol. $P = Fv = kv^2v$

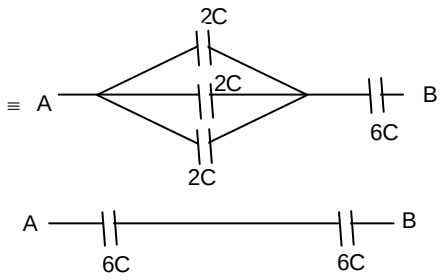
$\Rightarrow$ $P \propto v^3$

$\Rightarrow$ Ratio = 1 : 8



15. 00003.90

Sol. Equivalent circuit of the above figure can be drawn as $C_{AB} = 3C$



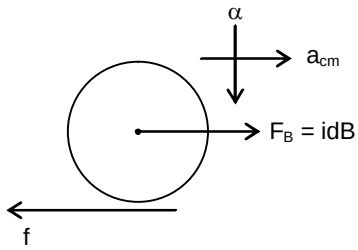

16. 00015.00

Sol. $V_B \cos\theta_B = V_A \cos\theta_A \Rightarrow V_B = 15\,m/s$

17. 00002.00

Sol. $f = \frac{F}{3} = \frac{idB}{3}$

$\Rightarrow$ $\frac{48 \times 0.5 \times 0.25}{3} = 2.00$




18. 00000.09

Sol. $\varepsilon = vB\ell = 3 \times 10^7 \times 10 \times 3 \times 10^{-10} = 9 \times 10^{-2} \approx 0.09$ V

# Chemistry

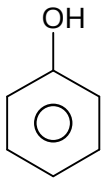
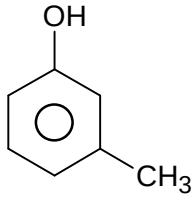
## PART – II

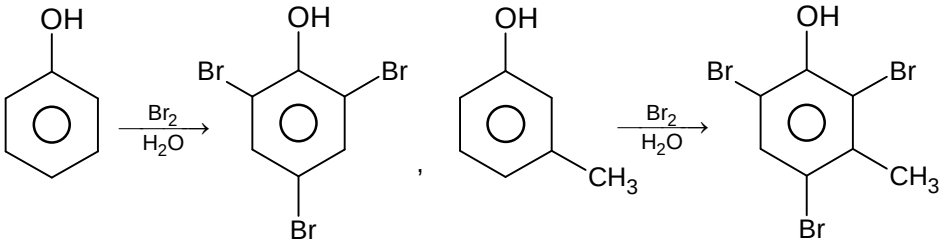
### SECTION – A

19. AB

Sol. The solubility of the group-2 sulphates decreases on moving down the group due to decrease in hydration energy.

20. AC

Sol. All the ortho and para positions of  and  are vacant and activated for electrophilic substitution reactions.




21. ABD

Sol. $H_3BO_3 \xrightarrow{-H_2O} HBO_2$

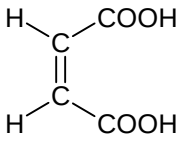
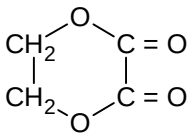
$2H_3BO_3 \xrightarrow{-3H_2O} B_2O_3$

$4H_3BO_3 \xrightarrow{-5H_2O} H_2B_4O_7$

$H_2B_2O_3$ is not an anhydride.

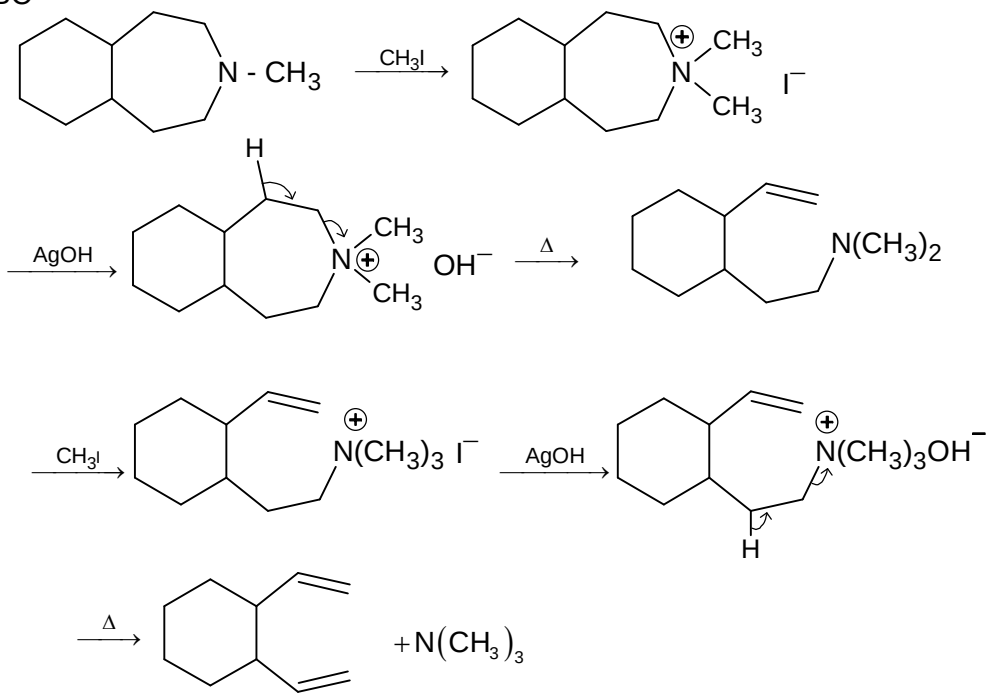
22. ABC

Sol. $C_4H_4O_4$ can be

 ,  and $H_2C = C(COOH)_2 \xrightarrow{\Delta} CH_2 = CHCOOH + CO_2$

23. BC

Sol.




24. ABC

Sol. These are facts.

25. A

Sol. (A) forms monosubstituted product, it is more reactive than benzene due to presence two activating $CH_3$ groups.

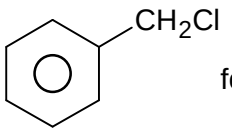
(B) forms monosubstituted product because one position is meta w.r.t electron withdrawing groups and para with respect to $CH_3$ group. It is less reactive than benzene due to two deactivating and one activating group.

(C) forms more than one monosubstituted product and is more reactive than benzene due to two activating OH groups. The OH groups exert –I and +R effect.

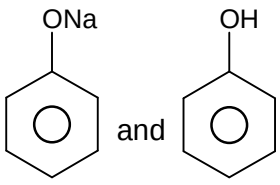
(D) forms more than one monosubstituted product and is less reactive than benzene. The Cl atoms exert –I as well as +R effect.

26. C

Sol.




forms stable benzylic carbocation which is stabilized by resonance




are more reactive than toluene towards bromination

27. B

Sol. Fact based.

28. D
Sol. Fact based

## SECTION – D

29. 00058.50
Sol. P is NaCl

30. 00055.80
Sol. $\Delta T_f = iK_f m = 3 \times 1.86 \times \text{No. of moles} \times \frac{1000}{W}$

or, $5 = 3 \times 1.86 \times 50 \times \frac{1000}{W} \Rightarrow w = 55.8$ kg

31. 00071.70
Sol. $CH_3CH_2CH_2CH_3 \xrightarrow[h\nu]{Cl_2} CH_3CH_2CH_2CH_2Cl + CH_3CH_2\underset{Cl}{\underset{|}{C}}H - CH_3$

$$\frac{\text{amount of } 1-\text{Chlorobutane}}{\text{amount of } 2-\text{Chlorobutane}} = \frac{6\times1}{4\times3.8} = \frac{6}{15.2}$$

$$\% \text{ of 2-chlorbutane} = \frac{15.2}{21.2}\times100 = 71.70\%$$

32. 00072.00
Sol. n double bonds dissociate to form 2n single bond.
$\Delta H$ for polymerization of 1 mol of ethylene

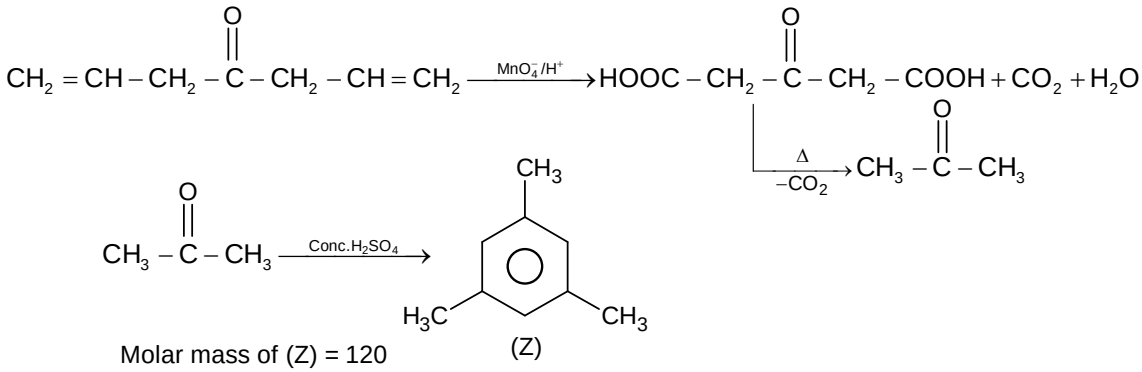
$$\Delta H = \frac{n}{n}\left(BE_{C=C}\right) - \frac{2n}{n}\left(BE_{C-C}\right)$$

$= +590 - 2 \times 331 = -72$ kJ mol$^{-1}$ $= -72.00$ kJ mol$^{-1}$

33. 00046.20
Sol. $t_{1/2} = \frac{0.693}{K} = \frac{0.693}{15\times10^{-3}} = 46.2\,s$

34. 00252.00
Sol. $X = (NH_4)_2Cr_2O_7$, $Y = Cr_2O_3$ & $Z = N_2$

35. 00120.00
Sol.




Molar mass of (Z) = 120

36. 00031.00

Sol. $2NO_3^- \longrightarrow N_2O_5$

Mole of ratio of $NO_3^-$ : $N_2O_5$ = 2 :1

$\therefore$ Mole of $NO_3^-$ in salt = 2 $\times$ Mole of $N_2O_5$ = $\frac{2\times1.08}{108} = 0.02$

$\therefore$ Weight of $NO_3^-$ in the salt = 0.02 $\times$ 62 = 1.24 g

% of $NO_3^-$ = $\frac{1.24}{4}\times100 = 31\%$

$\therefore$ X = 31.00

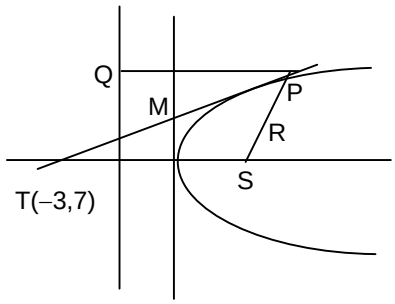
# Mathematics

## PART – III

### SECTION – A

37. ABCD

Sol. Equation of tangent at P is x+y=4




Clearly mirror image of R(–5,5) lies on line PQ.

Now mirror image R′ or R

$\Rightarrow \frac{\alpha'+5}{1} = \frac{\beta'-5}{1} = \frac{-2(-5+5-4)}{2} = 4$

$\Rightarrow (\alpha',\beta') \equiv (-1,9)$

Let PM cuts the axis at T; As M is midpoint of PT

$\Rightarrow$ T is (–3, 7)

We know that SP = ST and $\angle SMP = \frac{\pi}{2}$

Equation SP $\equiv y-3 = -\frac{1}{3}(x-1)$

$\Rightarrow x + 3y - 10 = 0$

Let $S \equiv (10-3\beta, \beta)$

Again TS || PQ $\Rightarrow \frac{\beta-7}{13-3\beta} = \frac{9-3}{-1-1} = -3$

$\therefore$ focus is (–2, 4)

38. ABCD

Sol. $f(x) = f(\alpha + \beta - x)$

$\Rightarrow f(\alpha) = f(\beta)$ and $g(\alpha) = g(\beta)$

$\int_\alpha^\beta f(x)g'(x)\,dx = f(x)\,g(x)\big|_\alpha^\beta - \int_\alpha^\beta f'(x)g(x)dx$

$2I = \int_\alpha^\beta f(x)g'(x) - \int_\alpha^\beta f'(x)g(x)dx$

$I = \frac{1}{2}\int_\alpha^\beta \left(f(x)\,g'(x) - f'(x)g(x)\right)dx$

39. AD

Sol. $h'(x) = f(x) \Rightarrow h''(x) = f'(x)$

$h(1) = 0, f(1) = f'(1) = h'(1) = h''(1) = g(1)$

$f(g(x)) = x$

$f'(g(x)).\ g'(x) = 1$

$f'(g(1)).\ g'(1) = 1 \Rightarrow g'(1) = 1$

$G(x) = x^2.g(x) - x.\ h(g(x))$

$G'(x) = x^2\ g'(x) + 2x.\ g(x) - h(g(x)) - x.\ h'(g(x)).\ g'(x)$

$= 2x.g(x) - h(g(x))$

$G''(x) = 2.g(x) + x.g'(x)$

$G'(1) = 2g(1) - h(g(1)) = 2g(1) - h(1) = 2$

$G''(1) = 2.g(1) + g'(1) = 3$

40. BD

Sol. Let one probability of choosing one integer k be $P(k)=\frac{\lambda}{k^4}$. ($\lambda$ is one constant of proportionality). Then,

$\sum_{k=1}^{2m}\frac{\lambda}{k^4}=1\Rightarrow\lambda\sum_{k=1}^{2m}\frac{1}{k^4}=1$

Let $x_1$ be the probability of choosing the odd number. Then

$x_1=\sum_{k=1}^{2m}P(2k-1)=\lambda\sum_{k=1}^{m}\frac{1}{(2k-1)^4}$

Also, $1-x_1=x_2=\sum_{k=1}^{m}P(2k)=\lambda\sum_{k=1}^{m}\frac{1}{(2k)^4}<\lambda\sum_{k=1}^{m}\frac{1}{(2k-1)^4}$

$\Rightarrow\ 1-x_1<x_1 \quad\Rightarrow\ x_1>\frac{1}{2} \quad\Rightarrow\ x_2<\frac{1}{2}$

41. BCD

Sol. $f''(x)\Rightarrow\begin{cases}2\,;\,x>0\\-2\,;\,x<0\end{cases}\Rightarrow$ not differentiable

$g(x)\Rightarrow\begin{cases}-x^2\cot^{-1}x;\quad x>0\\-x^2\cot^{-1}x;\quad x<0\\0\ ;x=0\end{cases}\Rightarrow$ twice differentiable

$h(x)=\sin^2x\ \rightarrow$ differentiable

$k'(x)=\begin{cases}4x^3.\cos\frac{1}{x}+x^2.\sin\frac{1}{x}\,;\,x\neq0\\0\,;\ x=0\end{cases}$

twice differentiable

42. AD

Sol. Using integration as sum of limit.

43. B

Sol. (A) $(x-h)^2+y^2=(h-1)^2$

$y^2=4x$

$(x-h)^2+4x=(h-1)^2$

Apply $\Delta=0$

(B) Find the common normal to

$y^2=4x,\ y^2=2x-6$

$y=m(x-3)-m-\frac{1}{2}m^3$ Apply $c=-2am-am^3$

$-4m-\frac{1}{2}m^3=-2m-m^3$

$\frac{m^3}{2}-2m=0$ $m=0,\ m=\pm2$

(C) Length of the semi latusrectum is the H.M between segments of the focal chord.

(D) Tangents at the ends of focal chord are perpendicular

44. A

Sol. Angles of the triangle are 30º, 40º, 105º

then $\frac{a}{\sin 30^\circ} = \frac{b}{\sin 45^\circ} = \frac{c}{\sin 105^\circ}$

$\Rightarrow \quad \frac{a}{1/2} = \frac{b}{1/\sqrt{2}} = \frac{c}{(\sqrt{3}+1)/2\sqrt{2}} = k'$

$\Rightarrow \quad \frac{a}{\sqrt{2}} = \frac{b}{2} = \frac{c}{\sqrt{3}+1} = k$

The smallest side $a = \sqrt{2}k$ and largest side $c = (\sqrt{3}+1)k$

If $a = 2, k = \sqrt{2}, c = \sqrt{6}+\sqrt{2}$

If $a = \sqrt{2}, k = 1, c = \sqrt{3}+1$

If $s = 3+\sqrt{2}+\sqrt{3}, k = 2, c = 2+2\sqrt{3}$

If $\Delta = \frac{\sqrt{3}-1}{4}$, then from $\Delta = \frac{1}{2}bc\sin A$

We have $\frac{\sqrt{3}-1}{4} = \frac{1}{2} \times 2k \times (\sqrt{3}+1)k \sin 30^\circ$

$\Rightarrow \quad k^2 = \frac{\sqrt{3}-1}{2(\sqrt{3}+1)} = \frac{(\sqrt{3}-1)^2}{4} = k = \frac{\sqrt{3}-1}{2}$ and

hence c = 1

45. D

Sol. Let $B_1, B_2, B_3, B_4$ are boys $F_1, F_2, F_3, F_4$ and $M_1, M_2, M_3, M_4$ are corresponding fathers and mothers. Let A; be an event when Bi, Fi, Mi are together in a group.

$n(A_1) = 3! \times 3!$ [Remaining 3 fathers can be grouped with any of 3 boys and similarly 3 mothers can be grouped with 3 boys separately]

Similarly $n(A_2), n(A_3), n(A_4) = 3! \times 3!$

$n(A_1 \cap A_2) = 2! \times 2!$

$n(A_1 \cap A_2 \cap A_3) = 1!1!$

$n(A_1 \cap A_2 \cap A_3 \cap A_4) = 1$

At least are Boy is grouped with his parents

$n(A_1 \cup A_2 \cup A_3 \cup A_4)$

$= \sum_{i=1}^{4} n(A_i) - \sum_{1=i<j}^{j=4} n(A_i \cap A_j) + \sum_{1=i<j<k}^{k=4} n(A_i \cap A_2 \cap A_3) - n(A_1 \cap A_2 \cap A_3 \cap A_4)$

$= {}^4C_1 \times 3! \times 3! - {}^4C_2 \times 2!2! + {}^4C_3 1! \times 1! - 1 = 123$

Grouping when any boy can be given any set of parents is $4! \times 4!$

(P) $n(\overline{A_1 \cup A_2 \cup A_3 \cup A_4}) = 4!4! - 123 = 576 - 123 = 453$

(Q) 2 boys selected four both parents in group in ${}^4C_2$ ways say $B_1$ and $B_2$ so $F_1B_1M_1$, $F_2B_2M_2$ are 2 groups remaining possibilities are

(i) $F_3B_3M_4$ and $F_4B_4M_3$

(ii) $F_4B_3M_3$ and $F_3B_4M_4$

(iii) $F_4B_3M_4$ and $F_3B_4M_3$ ${}^4C_2 \times 3 = 18$

(R) Boys are not selected for being particular

(S) $n(A_1 \cap A_2 \cap A_3 \cap A_4) = 1$

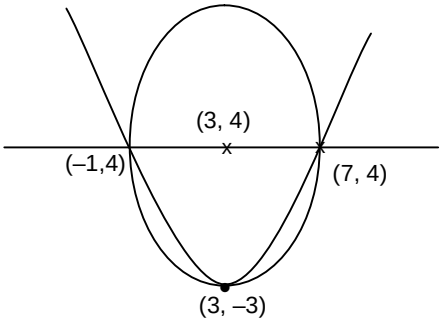
46. C

Sol. Basic definition of hyperbola and common tangent with circle.

## SECTION – D

47. 00005.00

Sol. Given ellipse $\frac{(x-3)^2}{4^2} + \frac{(y-4)^2}{7^2} = 1$ (vertical ellipse)




Parabola can be taken as

$(x-3)^2 = A(y+3)$

It passes through (-1, 4)

$\Rightarrow 16 = 7A \Rightarrow A = 16/7$

$\therefore$ parabola is $7(x-3)^2 = 16y + 48$

$\therefore 16y = 7(x-3)^2 - 48$

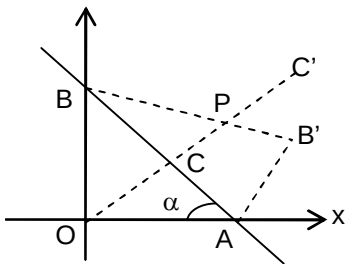
$\therefore A = 7, H = 3, K = 48$

$\therefore \frac{A}{7} + \frac{H}{3} + \frac{K}{16} = 5$

48. 00002.00

Sol. $A \equiv (1,0)$, $B = \equiv (0, \tan\alpha)$, $C \equiv \left(\frac{1}{2}, \frac{\tan\alpha}{2}\right)$, $B' = \left(\frac{2\tan^2\alpha}{1+\tan^2\alpha}, \frac{\tan^3\alpha - \tan\alpha}{1+\tan^2\alpha}\right)$

$\frac{\Delta ABB'}{\Delta BB'C} = 2$




49. 00028.00

Sol. If $a_1$ is mapped to 2, we have $^7C_5$ ways of mapping rest of the elements.
If $a_1$ is mapped to 3, we have $^6C_5$ ways of mapping rest of the elements.
If $a_1$ is mapped to 4, we have $^5C_5$ ways of mapping rest of the elements.
Hence total number of increasing functions = $^7C_5 + {}^6C_5 + {}^5C_5 = 28$.

50. 00007.00

Sol. $\left({}^{19}c_0(x^3)^{19} + {}^{19}c_1(px^2+2x-5)(x^3)^{18} + ....\right)$

$\left({}^8c_0x^{16} + {}^8c_1(qx-41)x^{14} + ....\right)\left({}^6c_0x^{24} + {}^6c_1(-x^3+x-1)x^{20} + .....\right)$

$= x^{97} + 391x^{96} + a_{95}x^{95} + ....$

Comparing the coefficient of $x^{96}$ we get

$19p + 8q - 6 = 391$

$\Rightarrow 19p + 8q = 397$ Let $q = 19\lambda + k$, $0 \le k < 19$

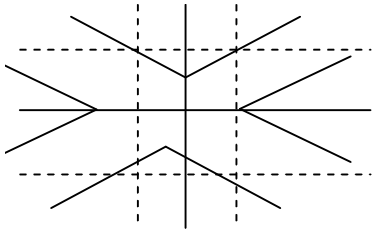
$p = \dfrac{397 - 8(19\lambda + k)}{19} = 21 - 8\lambda - 2\dfrac{(4k+1)}{19}$

$\dfrac{4k+1}{19}$ must be integer $\Rightarrow k = 14$, $p = 15 - 8\lambda$

For minimum positive value of p, $\lambda = 1 \Rightarrow p = 7$

51. 00002.00

Sol.



$x = \dfrac{-1}{2}$ $x = \dfrac{1}{2}$

52. 00002.00

Sol. $f'(x) = -2 - 3x^2 < 0 \ \forall \ x$

$\Rightarrow f(x)$ is decreasing

$f(f(x)) < f(-x)$

$f(x) > -x$

$30 - 2x - x^3 > -x$

$x^3 + x - 30 < 0$

$(x-3)(x^2+3x+10) < 0 \Rightarrow x < 3$

53. 00005.00

Sol. Let $\vec{d}.\vec{a} = \cos y\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = -\vec{d}.\left(\vec{b}+\vec{c}\right)$ [as $\vec{d}.\left(\vec{a}+\vec{b}+\vec{c}\right) = 0$]

$\Rightarrow \quad \cos y = -\dfrac{\vec{d}.\left(\vec{b}+\vec{c}\right)}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}$ .....(1)

similarly $\sin x = -\dfrac{\vec{d}.\left(\vec{b}+\vec{a}\right)}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}$ .....(2)

$$\Rightarrow 2 = -\frac{\vec{d}.\left(\vec{a}+\vec{c}\right)}{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]} \qquad \text{.....(3)}$$

Adding these we get $\sin x + \cos y + 2 = 0$

$\Rightarrow \sin x + \cos y = -2$

$\Rightarrow \sin x = -1, \cos y = -1$

$\Rightarrow x = (4n-1)\frac{\pi}{2},\ y = (2n+1)\pi$

Since we want minimum value of $x^2 + y^2$, so $x = -\frac{\pi}{2},\ y = \pi$

$\Rightarrow x^2 + y^2 = \frac{5\pi^2}{4} \Rightarrow \lambda = 5$

54. 00002.00

Sol. Ans. (2)

Put $x = t^2$